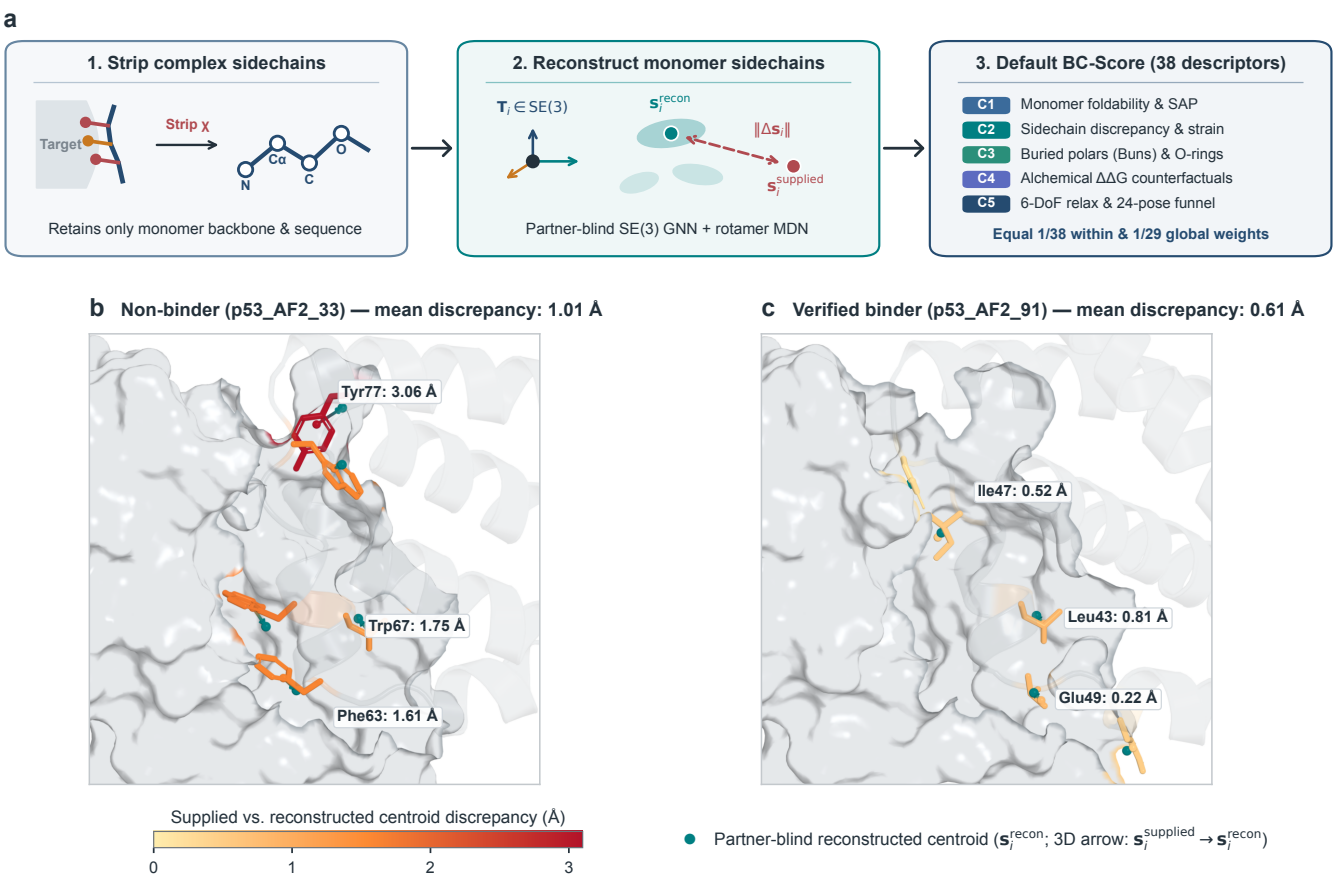

a
1. Strip complex sidechains
Target
Strip χ
Cα
O
N
C
Retains only monomer backbone & sequence
2. Reconstruct monomer sidechains
$T_i \in SE(3)$
s_i^{recon}
$\|\Delta s_i\|$
$s_i^{supplied}$
Partner-blind SE(3) GNN + rotamer MDN
3. Default BC-Score (38 descriptors)
C1 Monomer foldability & SAP
C2 Sidechain discrepancy & strain
C3 Buried polars (Buns) & O-rings
C4 Alchemical ΔΔG counterfactuals
C5 6-DoF relax & 24-pose funnel
Equal 1/38 within & 1/29 global weights
b Non-binder (p53_AF2_33) — mean discrepancy: 1.01 Å
Tyr77: 3.06 Å
Trp67: 1.75 Å
Phe63: 1.61 Å
c Verified binder (p53_AF2_91) — mean discrepancy: 0.61 Å
Ile47: 0.52 Å
Leu43: 0.81 Å
Glu49: 0.22 Å
Supplied vs. reconstructed centroid discrepancy (Å)
0
1
2
3
Partner-blind reconstructed centroid (s_i^{recon}; 3D arrow: $s_i^{supplied} \rightarrow s_i^{recon}$)